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🐙 [Marty99-dev](https://github.com/Marty99-dev)

Mathias Marty

Education

- 2026–today **PhD in Computer Science**, *EPFL*, Lausanne
- 2022–2024 **Master of Science - Cyber Security**, *EPFL/ETHZ*, Lausanne/Zürich, GPA: 5.34/6
- 2021–2022 **HES–EPFL transition program**, *EPFL*, Lausanne, GPA: 5.12/6
- 2018–2021 **Bachelor of Science - Computer Science**, *HES/SO Haute Ecole Arc*, Neuchâtel, GPA: 5.24/6
- 2015–2018 **Federal Diploma of Vocational Education and Training (Computer Science) and Professional Baccalaureate**, *CPNE*, Neuchâtel

Publications

Under review / In preparation

- [1] Jonathan Bootle, Alessandro Chiesa, and Mathias Marty. Round-by-round knowledge soundness for linear-time iops. 2026. Manuscript in preparation.
- [2] Iftach Haitner, Mathias Marty, and Eylon Yogev. Lower bound for the size of hash based signatures. 2026. Manuscript in preparation.
- [3] Mathias Marty. Counting successor-closed subsets of functional digraphs, 2026. [\[link\]](#) .

Published / Accepted Papers

- [4] Antoine Lestrade, Mathias Marty, Artan Sadiku, Christophe Muller, Joep Neijt, Yann Voumard, and Stéphane Gobron. Real-time renderings of multidimensional massive datacubes on jupyter notebook. In Liang-Yee Cheng, editor, *Proceedings of the 20th International Conference on Geometry and Graphics*, ICGG '22, pages 685–696. Springer International Publishing, 2023. [\[link\]](#) .
- [5] Mathias Marty, Antoine Lestrade, Artan Sadiku, Christophe Muller, Joep Neijt, Yann Voumard, and Stéphane Gobron. Implicit curves: From discrete extraction to applied formalism. In *Proceedings of the 20th International Conference on Geometry and Graphics*, ICGG '22, pages 717–728, 2022. [\[link\]](#) .
- [6] Christophe Muller, Antoine Lestrade, Mathias Marty, Artan Sadiku, Joep Neijt, Yann Voumard, and Stéphane Gobron. Earth observation datacubes multi-visualization toolbox. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-4/W1-2022, 2022. [\[link\]](#) .

Research projects

- 2024–present **Independent Research**, with Prof. Alessandro Chiesa and Dr. Jonathan Bootle
Round-by-Round Knowledge Soundness for Linear-Time IOPs
- 2025–present **Independent Research**, with Prof. Eylon Yogev and Prof. Iftach Haitner
Implementing a PoC in Rust and lower bounds for hash-based signatures
- 2024 **Master Project**, supervised by Prof. Alessandro Chiesa, EPFL, grade: 5.75/6
On State-Restoration Knowledge Soundness from Special Soundness ([pdf](#), [presentation](#))
- 2023 **Semester Project**, supervised by Prof. Serge Vaudenay, EPFL, grade: 5.75/6
Code-based Cryptography in the Lee metric ([pdf](#), [presentation](#))
- 2022 **Formal Verification Course Project**, EPFL, not graded
Prove some RSA security properties with the proof assistant EasyCrypt
- 2021 **Bachelor Project**, supervised by Prof. Stéphane Gobron, HES/SO Haute Ecole
Arc, grade: 6/6
DataCube: Implicit Curves ([pdf \(french only\)](#))

Skills

- Practical In addition to my theoretical skills, I acquired many practical skills.
- **Programming:** Rust, Assembly, C, VHDL, C++, Scala & Stainless, Java, C#, JavaScript, WebGL/GLSL, SQL, Python/Sage, Kotlin
 - **Software and tools:** Git, Linux, EasyCrypt, GDB, CUDA, $\text{\LaTeX}$, QEMU, RTOS, embedded devices, Docker, American Fuzzy Lop
 - **Soft skills:** Writing technical reports and papers, making presentations, teaching, ability to adapt to different learning levels and needs, communication
- Related Courses Cryptography and Security (5), Advanced Cryptography (5.25), Number Theory in Cryptography (5.5), Foundations of Probabilistic Proofs (5), Computational Complexity (5.25), Computer Architecture (5), Operating Systems (5.5), Computer Networks (5.75), Algorithms (5), Advanced Algorithms (5.25), Formal Verification (5), Gödel and Recursivity (5.75), Advanced Encryption Schemes, Zero-Knowledge Proofs...

Experience

- 2026 **Software Engineering Internship**, Corintis, Lausanne
Software engineer at Corintis, developing software for chip-level optimized cooling systems
- 2025 **Programmer**, Prof. Eylon Yogev, 50% work for Prof. Eylon Yogev
Implementing cryptographic primitives and proofs of concepts
- 2023 **Summer internship**, Swiss Center for Electronics and Microtechnology (CSEM), Neuchâtel, supervised by Dr. Damian Vizár
Embedded fuzzing testbench of a RTOS with HALucinator *evaluation: Excellent*
- 2018–present **Private tutoring**
Tutored students across a wide academic range, from elementary to University level (Also teaching on [Superprof account](#))

Languages

French (native), English (fluent; TOEFL 102/120 in 2024, FCE in 2018)